Discrete Optimization of Geometric Compactness: A Dual Heuristic Approach to 2D Tiling

Study on the automatic assembly of connected blocks — Karl Garrido Anderson

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Abstract

This article presents a methodology for the automated assembly of discrete geometric shapes on a grid. Faced with the NP-complete nature of the tiling problem, we present three progressive heuristics based on block morphology. The first heuristic H_1 relies on a greedy selection of maximal-order vertices, while the second, H_2, exploits a structural adaptation metric between vertices. The third heuristic H_3, a *locally exhaustive* greedy brute-force method, serves as an empirical benchmark to assess the quality of solutions on moderate-size instances. The joint use of the compactness index and a morphological resemblance metric makes it possible to reach dense configurations and to finely quantify the gap between heuristic solutions and the theoretical optimum. Reading this document provides all the keys to understanding the model, from vertex topology to fusion logic.

0. The author certifies that all of the mathematical reasoning, the topological model and the algorithmics presented in this document are the result of original personal research. The only exception is the NP-complete classification of the tiling problem [1], an established result of the literature.

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1 Introduction

The problem of automatically assembling blocks on a regular grid is a classical yet deep challenge in algorithmic geometry and combinatorial optimization. This study is part of the search for efficient solutions to two-dimensional tiling, where the arrangement of pieces must satisfy strict structural constraints while maximizing the use of the available space.

1.1 Problem statement and objectives

The central objective of this work is to maximize the **compactness** of a final assembly built from a set of initial connected blocks. In this framework, compactness is defined as the ratio between the total area and the exposed perimeter of the shape (see Definition 2.4). An optimal assembly therefore tends toward a dense structure, minimizing interstitial voids and maximizing the useful surface relative to the contour.

1.2 Complexity and approach

The tiling and placement optimization problem is known to be **NP-complete** [1]. This complexity makes the use of exact algorithms prohibitive for large instances, the time complexity of an exhaustive search being of the order of $O(c^n \cdot n!)$. Consequently, we adopt an approach based on the development and analysis of three successive heuristics, of increasing complexity and quality:

1. H_1 — **Greedy maximization:** Selection of maximal-order vertices, quasi-linear complexity $O(V + N)$.
2. H_2 — **Optimization by gap metric:** Exploration of the space of tip–hollow pairs with structural filtering, quadratic complexity $O(V^2 \cdot N)$.
3. H_3 — **Greedy brute force:** Exhaustive evaluation of all fusion configurations between the two selected blocks, serving as an empirical benchmark for moderate-size instances.

1.3 Applications and industrial stakes

Although this study focuses on a theoretical and algorithmic formalization, the practical benefits of compactness optimization are considerable. The algorithms developed here find a direct application in fields of resource storage under strong spatial constraints, such as:

- **Logistics and warehousing:** Optimization of the surface occupied by irregularly shaped goods in order to maximize warehouse capacity and reduce infrastructure costs.
- **Industrial packaging:** Reduction of void volumes in packaging to lower transport costs and carbon footprint.
- **Material resource management:** Spatial organization of components or raw materials whose physical bulk limits operational capacity.

1.4 Main contributions

The major contributions of this article revolve around four axes:

- The formalization of a **compactness metric** suited to discrete environments, and the proof that the square maximizes its value among rectangles.
- The development of a **topological model of vertices** distinguishing tips and hollows to guide geometric fusions.

- The design of an **interlocking strategy** preventing overlaps and guaranteeing admissible assemblies.
- An in-depth experimental study comparing the algorithmic performance of the heuristics and the morphology of the solutions obtained, enriched by a weighted structural resemblance metric.

2 Framework and fundamental definitions

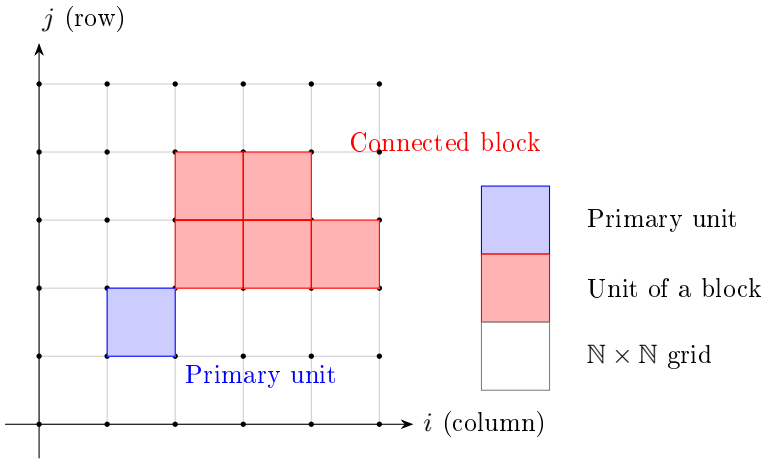
2.1 Definitions

We define here the basic mathematical objects that will be used throughout this article. These definitions form the formal foundation indispensable to understanding the heuristics.

Definition 2.1 (Discrete space)**.** *The workspace E is defined by the discrete grid $\mathbb{N} \times \mathbb{N}$. Each point $(i, j) \in E$ represents the lower-left corner of a primary unit.*

Definition 2.2 (Primary unit)**.** *The primary unit is the square of side 1 whose lower-left corner is a point of E. It constitutes the irreducible atomic element of every block.*

Definition 2.3 (Block)**.** *A **block** is a finite and connected set of primary units, where connectivity is defined by sharing at least one face (4-connectivity). The set of all blocks is denoted $\mathcal{B}$.*



2.2 Compactness theory

The compactness index $c(f)$ is our main metric for evaluating the quality of a shape. Intuitively, this index measures the area/perimeter ratio: the more "round" or compact a shape is, the higher this index.

Definition 2.4 (Compactness index)**.** *For a block $f \in \mathcal{B}$ of area A (number of primary units) and exposed perimeter P (number of boundary edges), the compactness index is defined by:*

$$c(f) = \frac{A}{P}$$

Remark 2.1. *For an $m \times n$ rectangle, we have $A = mn$ and $P = 2(m + n)$, which gives $c = \frac{mn}{2(m+n)}$. For an $n \times n$ square, we obtain $c = \frac{n}{4}$, which grows linearly with n, confirming the intuition that large square shapes are the most compact.*

Proposition 2.1. *Let an $m \times n$ grid belong to E. Among all connected blocks whose bounding box is $m \times n$, the full rectangle of size $m \times n$ is the one that maximizes the compactness index.*

Proof. See Annex 8.2.4. □

Proposition 2.2. *Let $n \in \mathbb{N}^*$. Among all full rectangles of dimensions $a \times b$ such that $a \times b = n^2$, the square of side n is the one that maximizes the compactness index.*

Proof. See Annex 8.2.4. □

Corollary 2.1. *In the discrete space E, among all full rectangles of fixed area n^2, the square of side n maximizes the compactness index. This result justifies that the fusion heuristics naturally tend to produce square or quasi-square assemblies.*

Proposition 2.3. *Let $n \in \mathbb{N}^*$. For every block $b \in \mathcal{B}$ of area $A_b \leq n^2$, the full square of side n (denoted S_n) is the most compact block. In other words:*

$$\forall b \in \mathcal{B}, \quad A_b \leq n^2 \implies c(b) \leq c(S_n) = \frac{n}{4} \tag{1}$$

Proof. See Annex 8.2.4. □

Remark 2.2. *The extension of this result to non-rectangular connected shapes (general polyominoes) is an open question in the discrete setting.*

3 Vertex topology

Before defining the assembly heuristics, it is necessary to introduce a fine topological structure of block contours. It is this local analysis of vertices that will make it possible to guide fusion decisions in a geometrically coherent way.

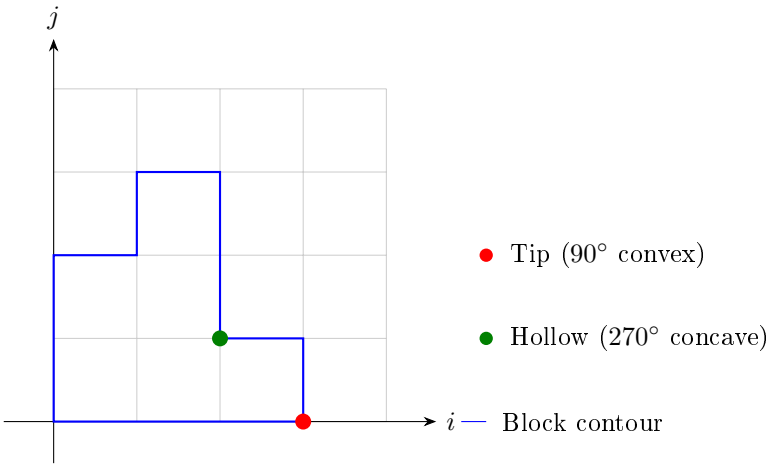
3.1 Vertex types

The contour of a block is defined by two types of critical points, characterized by their interior angle:

— **Tip (p):** *Convex* vertex with an interior angle of 90°. It is a protrusion of the block, a point where the contour "points" outward.

— **Hollow (q):** *Concave* vertex with an interior angle of 270°. It is a notch of the block, a point where the contour turns inward.

Remark 3.1. *Only* 90° *and* 270° *angles exist on a discrete contour (orthogonal grid).* 180° *angles correspond to straight segments and do not constitute critical vertices.*



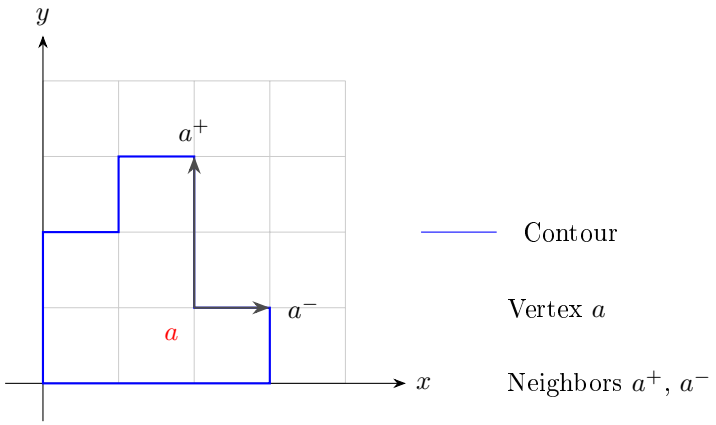
3.2 Formal representation of a vertex

For the rest of our study, we formalize the topological structure of critical points (tips and hollows). Let a be a vertex of the contour of a block. We define a^+ as the direct neighbor of a located in the adjacent "left" space relative to the orientation of the contour, and a^- as its direct neighbor on the "right".

Each vertex a is represented by a triplet made up of its Cartesian coordinates and its adjacent segments:

$$a = \big((a_x, a_y),\ a^+,\ a^-\big) \tag{2}$$

where $(a_x, a_y) \in \mathbb{N}^2$ represents the coordinates of the point in the discrete space E.



3.3 Order of a vertex

Definition 3.1 (Order of a vertex). *The order $\omega(a)$ of a vertex a is the pair (d_1, d_2) where d_1 and d_2 are the lengths of the boundary segments adjacent to a in the two orthogonal directions. The product $d_1 \cdot d_2$ measures the* local area of influence *of the vertex, that is, the surface that an interlocking can cover at this vertex.*

3.4 Admissibility conditions for an interlocking

To merge two blocks f_1 and f_2, one seeks to fit a tip p of f_1 into a hollow q of f_2 (or conversely). The admissibility of this interlocking depends on the topology of the immediate neighbors of p and q.

Proposition 3.1 (Local overlap conditions). *Let p be a tip of f_1 and q a hollow of f_2, with their respective neighbors p^-, p^+ and q^-, q^+. The interlocking of p into q gives rise to four cases depending on the nature of the neighbors:*

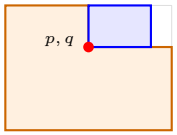
Case 1 — Bilateral convexity. *If $\{p^-, p^+\}$ are tips and $\{q^-, q^+\}$ are tips: the interlocking is trivially admissible. No additional distance constraint is required, the contours fit together perfectly.*

Case 2 — Protrusion of f_1. *If $\{p^-, p^+\}$ are tips but one of the neighbors of q (for example q^+) is a hollow: there is a protrusion of f_1 on the side of q^+. Condition: $\min(d(p^-, p), d(p, p^+)) \leq d(q, q^+)$ and the space surrounding q^+ must be free to accommodate the protrusion.*

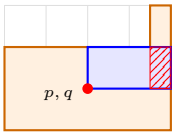
Case 3 — Protrusion of f_2. *If one of the neighbors of p (for example p^+) is a hollow while $\{q^-, q^+\}$ are hollows: the block f_2 presents a protrusion toward p^+. Condition: $d(p, p^+) \leq \max(d(q^-, q), d(q, q^+))$ and the internal space of f_1 must allow the reception of the segment of f_2.*

Case 4 — Bilateral protrusion. *If p has a hollow neighbor and q has a tip neighbor: the conditions of Cases 2 and 3 must be satisfied* simultaneously *to guarantee the absence of intersection.*

Proof. See Annex 8.2.4. □



Case 1: Admissible



Case 2: Inadmissible

Proposition 3.2 (Interchangeability by factorial decomposition)**.** *Let an admissible interlocking between a tip p of order $\omega = (d_1, d_2)$ and a hollow q exhibiting an axial symmetry along one of the orthogonal axes of the discrete plane. Every tip p' of order $\omega' = (d'_1, d'_2)$ satisfying $d_1 \cdot d_2 = d'_1 \cdot d'_2$ is also admissible with respect to q, provided that the initial translation system remains consistent with the new decomposition.*

Remark 3.2. *The axial-symmetry hypothesis on the hollow q is essential: it guarantees that the redistribution of lengths $(d_1, d_2) \to (d'_1, d'_2)$ preserves the contact surface without creating an overlap. In practice, this condition is satisfied for the majority of hollows encountered on blocks of low complexity (3 to 8 cells).*

Remark 3.3. *This result is important in practice: it means that two vertices with the same "area of influence" (same product $d_1 d_2$) are interchangeable in an interlocking under the aforementioned conditions. This follows from the decomposition into prime factors: if $d_1 \cdot d_2 = d'_1 \cdot d'_2$, the two tips share the same "geometric signature" with respect to the hollow q.*

Proof. See Annex 8.2.4. □

4 Unified algorithmic framework

Before detailing the individual heuristics, we present here the general algorithmic framework within which H_1, H_2 and H_3 fit. The three approaches share the same iterative greedy structure; they differ only by the function $h(f_1, f_2)$ used to select the best interlocking at each step (detailed in Sections 5 and 6).

Algorithm 1 Greedy Fusion Process (unified framework $H_1/H_2/H_3$)

Input: List L of blocks.
Initialization: Sort L by decreasing $c(f)$.
while $|L| > 1$ **do**
    Select $f_1 \leftarrow L[0]$ and $f_2 \leftarrow L[1]$.
    if f_1 and f_2 have no hollow **then**
        Merge the facets whose lengths are closest (fusion by surface).
    else
        Compute the candidate fusion via $h(f_1, f_2)$ (according to the chosen heuristic among H_1, H_2, H_3 — see Sections 5 and 6).
        if Overlap detected **then**
            Fallback strategy: apply a rotation or select a lower-ranked candidate.
        else
            Validate the fusion: $f_{res} \leftarrow f_1 \cup f_2$.
            Remove f_1 and f_2 from L, insert f_{res}.
            Re-sort L by decreasing $c(f)$.
Output: The final shape $L[0]$.

Remark 4.1. *The algorithm terminates in at most $n-1$ iterations (one fusion per step), where n is the number of initial blocks. Termination is guaranteed by the fallback mechanism of H_1 and H_2, which ensures that an admissible fusion is always found.*

5 Assembly heuristics: H_1 and H_2

The transition from topological analysis to the actual assembly of blocks requires rigorous decision criteria. The objective is to determine, at each step, which pair of blocks to merge and according to which geometric configuration, in order to minimize the overall perimeter. We explore here two distinct approaches: the first, H_1, relies on a purely quantitative view of compactness, while the second, H_2, favors the structural complementarity of shapes through the analysis of their critical points.

5.1 Heuristic 1: Compactness maximization (H_1)

5.1.1 Principle

This approach favors the aggregation of blocks to minimize the total exposed perimeter. By systematically selecting the fusions that immediately increase the index $c(f)$, this method acts as a classical greedy algorithm.

Structural limitation. H_1 may fill voids sub-optimally if the shapes are poorly suited, creating geometric "locks" that prevent any later fusion. Indeed, by considering only maximal-order vertices, the algorithm risks neglecting configurations that are locally less compact but globally more favorable.

5.1.2 Selection of candidate vertices

In order to favor the vertices with the greatest potential for geometric spreading, we define the index sets I_p (candidate tips) and I_q (candidate hollows) as the indices maximizing the local area of influence:

$$I_p = \{\, i \mid p_i(0) \cdot p_i(1) = \max_j \big(p_j(0) \cdot p_j(1)\big)\} \tag{3}$$

$$I_q = \{\, i \mid q_i(0) \cdot q_i(1) = \max_j \big(q_j(0) \cdot q_j(1)\big)\} \tag{4}$$

where $p_i(0)$ and $p_i(1)$ denote respectively the two components of the order of the i-th vertex (lengths of the adjacent segments in the two orthogonal directions).

5.1.3 Fusion protocol

The fusion strategy follows an iterative and stochastic protocol:

— **Selection:** An index $i \in I_p$ and an index $k \in I_q$ are drawn at random (drawing without replacement within the current step) to attempt a fusion between the tip p_i and the hollow q_k.

— **Failure handling:** If the fusion fails (overlap detected or topological constraint violated according to Proposition 3.1), the algorithm selects a new pair within the current sets, until all available pairs are exhausted.

— **Fallback strategy:** In the case where the sets I_p or I_q are exhausted without success, we widen the search to lower-ranked candidates. The sets are recomputed using the value immediately below the previous maximum:

$$\text{New target} = \max_j \big(p_j(0) \cdot p_j(1)\big) - 1 \quad \text{or} \quad \max_j \big(q_j(0) \cdot q_j(1)\big) - 1$$

5.1.4 Modeling the time complexity of H_1

The heuristic H_1 is of the restrictive greedy type. It considers only maximal-order vertices (the most prominent protrusions and the deepest hollows).

— **Analysis:** For each rotation (four in number), the algorithm identifies the maximal-order vertex in $O(V)$, where V is the number of vertices. It then attempts only a single fusion per case.

— **Complexity:** The complexity is dominated by the extraction of the vertices and the verification of the single fusion:

$$T(H_1) = O(V + N)$$

where V is the number of vertices and N the cost of the non-overlap test. H_1 is therefore quasi-linear with respect to the size of the shapes.

5.2 Heuristic 2: Geometric adaptation by gap metric (H_2)

5.2.1 Principle

For increased precision, H_2 introduces an adaptation metric between all admissible pairs (tip p_i, hollow q_i). Unlike H_1, which considers only extreme vertices, H_2 explores the complete space of possible combinations before selecting the most promising one according to a double criterion of gap minimization and contact surface maximization.

5.2.2 Definition of the set I

We define the set I of pairs minimizing the dimensional gap on at least one of the two axes:

$$I = \{\, i \mid (p_i - q_i)(0) = \min_j (p_j - q_j)(0) \;\vee\; (p_i - q_i)(1) = \min_j (p_j - q_j)(1) \}$$

Intuitively, I gathers the (tip, hollow) pairs whose dimensions are closest on at least one dimension. The use of the logical connective $\vee$ (inclusive OR) is deliberate: it produces a potentially large set I, thus guaranteeing that geometric diversity is preserved at this stage. The filtering is then tightened by the set I'. A simultaneous minimization criterion on both axes ($\wedge$) would be more restrictive but would reduce recall and increase the risk of obtaining no exploitable candidate.

5.2.3 Refinement to the set I'

This set is then refined to obtain I', by maximizing the potential contact surface among the candidates of I:

$$I' = \{\, i \in I \mid p_i(0)\, q_i(0) = \max_j \big(p_j(0)\, q_j(0)\big) \;\vee\; p_i(1)\, q_i(1) = \max_j \big(p_j(1)\, q_j(1)\big) \}$$

I' selects, among the best dimensionally adapted pairs, those that maximize the local area of the interlocking. It is the product $p_i(k) \cdot q_i(k)$ that measures the contact surface in direction k.

5.2.4 Fallback strategies

Analogously to H_1, we apply a systematic fallback strategy when the optimal criteria cannot be satisfied.

Degradation of the refinement criterion (I'). If the set I' is empty, the algorithm widens its search by accepting contact surfaces of immediately lower rank:

$$I'_{relax} = \{\, i \in I \mid \big(p_i(0)\, q_i(0) = \max_j (p_j(0)\, q_j(0)) - 1\big) \;\vee\; \big(p_i(1)\, q_i(1) = \max_j (p_j(1)\, q_j(1)) - 1\big) \} \quad (5)$$

Degradation of the gap-minimality criterion (I). If the set I is empty or if no fusion has been validated after relaxing I', the algorithm allows a larger gap:

$$I_{relax} = \{\, i \mid \big((p_i - q_i)(0) = \min_j (p_j - q_j)(0) + 1\big) \ \vee \ \big((p_i - q_i)(1) = \min_j (p_j - q_j)(1) + 1\big)\} \quad (6)$$

Remark 5.1. *This approach by incrementation ($\min +1$) or decrementation ($\max -1$) guarantees that the heuristic is never stuck in a state without a solution. The algorithm thus degrades the theoretical quality of the fusion in a controlled and progressive manner until a geometrically admissible interlocking is found. This mechanism constitutes a completeness guarantee for H_2.*

5.2.5 Modeling the time complexity of H_2

The heuristic H_2 is much more exhaustive than H_1 because it first generates the set of all possible pairs before restricting by the sets I and I'.

— **Analysis:** The first step generates the Cartesian product of the vertices of f_1 and f_2, producing $O(V_1 \cdot V_2)$ candidates. For each pair, a non-overlap test of cost N is performed.

— **Complexity:** With V_1 and V_2 vertices respectively,

$$T(H_2) = O(V^2 \cdot N)$$

This quadratic complexity explains why H_2 is slower than H_1, while being more geometrically precise.

5.3 Experimental protocol

Performance measurements are obtained according to the following protocol.

Execution environment. Measurements are carried out on Python 3.12 (Kaggle, CPU only). Times are measured via the `time.perf_counter` module. Each measurement covers the entire assembly process (generation excluded).

Note on sample size. With 5 repetitions per configuration, the high standard deviations observed (notably for H_2 at $n = 20$, ± 3397 ms) reflect the sensitivity to the topology of the instances. These results are presented as an illustration of asymptotic trends.

Polyomino generation. The cellular growth process follows four iterative steps guaranteeing the connectivity (4-connectivity) of each generated block:

1. **Initialization:** A first pixel (cell) is placed at the origin $(0, 0)$.
2. **Exploration:** The algorithm identifies the set of empty cells adjacent (4-connectivity) to the current shape.
3. **Expansion:** One cell is chosen at random among these candidates and added to the block.
4. **Stopping condition:** The cycle repeats until the number of cells reaches the target size $k \in [\![\min, \max]\!]$.

5.4 Results — Performance analysis of H_1 and H_2

5.4.1 Comparison of time complexity

This section analyzes the impact of algorithmic choices on the overall assembly time by comparing the simplified approach (H_1) and the analytical approach (H_2).

Theoretical analysis. From a theoretical standpoint, the performance gap is explained by the nature of the candidate search. While H_1 is limited to extreme vertices ($O(V)$), H_2 explores a quadratic space $O(V^2)$ before filtering. Figure 1 illustrates this gap.

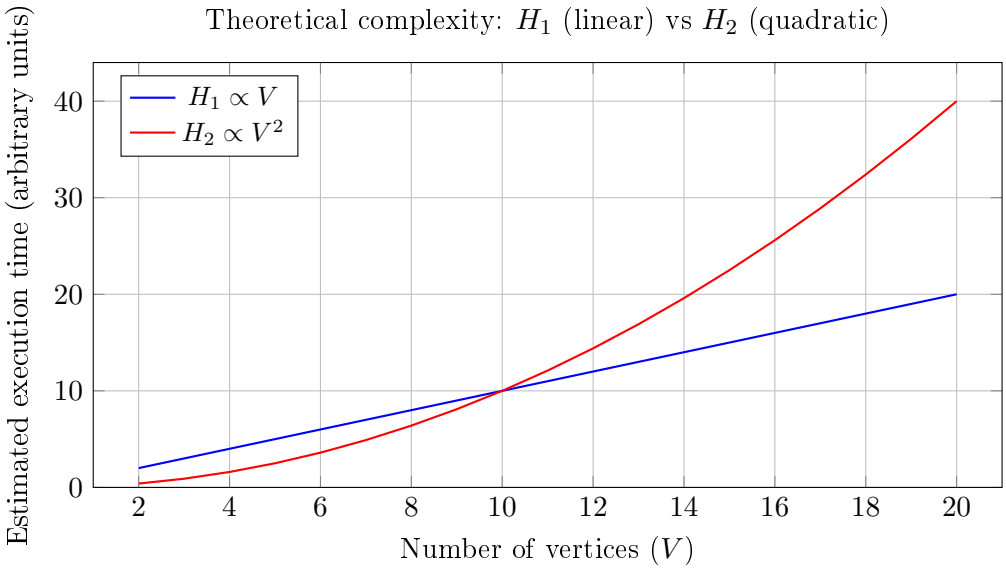


FIGURE 1 – Expected theoretical gap between direct selection (H_1, linear) and exhaustive analysis (H_2, quadratic).

Experimental results. The tests were carried out on instances ranging from $n = 2$ to $n = 24$ blocks, in accordance with the protocol described in Section **??**. Table 1 presents the mean computation times and their standard deviations.

Size (n)	Time H_1 (ms)	Time H_2 (ms)	Factor H_2/H_1
2	0.72 ± 0.20	0.86 ± 0.36	1.19
4	29.69 ± 22.41	34.46 ± 24.48	1.16
8	574.32 ± 288.69	$1{,}208.99 \pm 404.25$	2.10
12	$1{,}648.46 \pm 113.09$	$4{,}507.56 \pm 376.31$	2.73
16	$2{,}964.16 \pm 198.12$	$8{,}645.45 \pm 1{,}175.30$	2.91
20	$4{,}320.25 \pm 405.69$	$15{,}155.58 \pm 3{,}397.93$	3.50
24	$5{,}714.80 \pm 465.64$	$22{,}132.68 \pm 1{,}957.50$	3.87

TABLE 1 – Experimental measurements of execution time for H_1 and H_2 (5 trials per size, seeds 42–46, piece size: 3–8 cells).

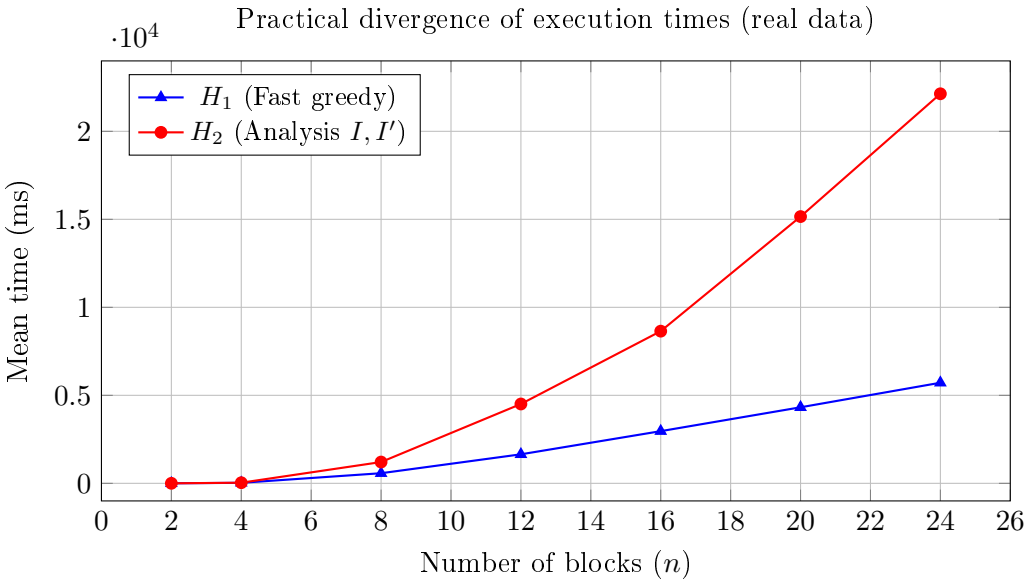


FIGURE 2 – Observed divergence of the execution times of H_1 and H_2. The quadratic nature of H_2 is clearly visible from $n = 8$.

5.4.2 Analysis of the time divergence

Observing the results of Table 1 allows us to draw three fundamental conclusions:

— **Complexity explosion of H_2:** While H_1 keeps a stable progression, H_2 undergoes an acceleration of its computation time that is characteristic of quadratic growth. At $n = 24$, H_2 is nearly **4 times slower** than H_1.

— **Geometric sensitivity:** The instability of the standard deviations for H_2 (for example ± 3397 ms for $n = 20$) suggests that it is extremely sensitive to the topology of the blocks: the more the number of vertices generated by the fusions increases, the more costly the $V_1 \times V_2$ space to explore becomes.

— **Precision/speed trade-off:** H_1 is ideal for contexts requiring a fast response. However, its effectiveness is limited by its geometric "blindness" — it examines only a fraction of the possible configurations — whereas H_2 invests time to guarantee structurally denser interlockings.

5.4.3 Comparison of the evolution of the compactness of H_1 and H_2

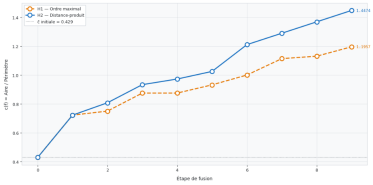


FIGURE 3 – 10 iterations

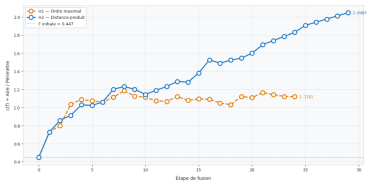


FIGURE 4 – 30 iterations

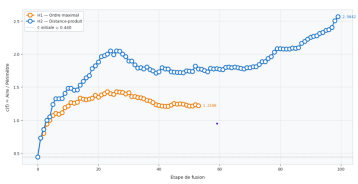


FIGURE 5 – 100 iterations

The analysis of these three graphs allows us to draw three main conclusions:

— **Superiority of H_2:** The heuristic H_2 proves clearly more effective than H_1 in terms of final compactness. This result is consistent: beyond the simple maximization of the surface covered during fusion, H_2 optimizes the structural fit by ensuring that the contours of the first piece match those of the second as closely as possible.

— **Robustness across instance sizes:** This efficiency is constant, both for small instances ($n \in \{10, 30\}$) and for the largest ones ($n = 100$). The performance gap widens as the assembly process unfolds.

— **Premature stop of H_1:** An abrupt interruption of the heuristic H_1 is observed on large instances. This phenomenon is explained by the exploration of a single branch: the algorithm gets trapped in a local optimum where any subsequent fusion inevitably causes an overlap, whatever the (tip, hollow) pairs considered.

These observations directly motivate the introduction of a *beam search* mechanism to avoid local optima, presented below.

5.4.4 Adding *Beam Search* to H_1 and H_2

Beam search is a beam-based exploration technique: instead of keeping only a single branch at each step (pure greedy strategy), the k best branches are maintained in parallel.

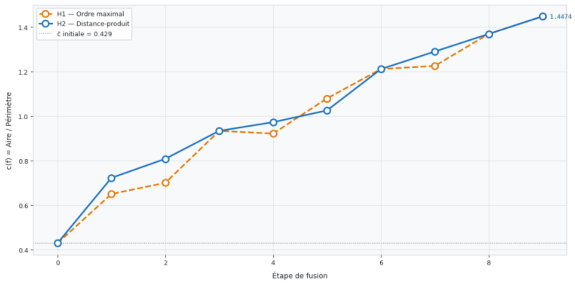


FIGURE 6 – Evolution of compactness with beam search, 10 iterations, $k = 10$ branches

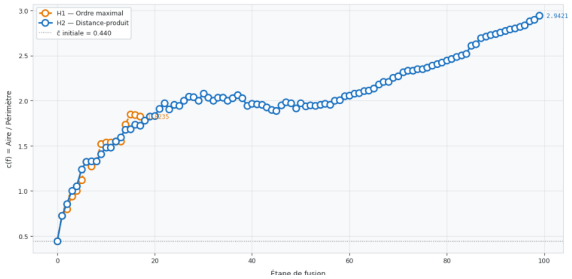


FIGURE 7 – Evolution of compactness with beam search, 100 iterations, $k = 10$ branches

As expected, the integration of previously discarded solutions allows H_1 to reach a level of compactness equivalent to that of H_2. However, this approach runs into a major complexity constraint: taking side solutions into account leads to an exponential growth in the number of branches to explore. On large instances, not only does the execution time grow prohibitively, but the risk of generating no complete final solution increases, as the process is interrupted as soon as the critical branch threshold is reached.

5.4.5 Analysis of the influence of the number of branches on H_1 and H_2

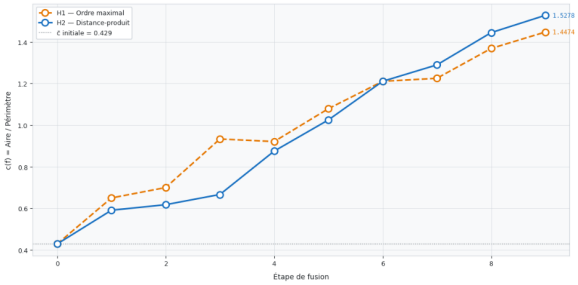


FIGURE 8 – Evolution of compactness with beam search, 10 iterations, $k = 100$ branches

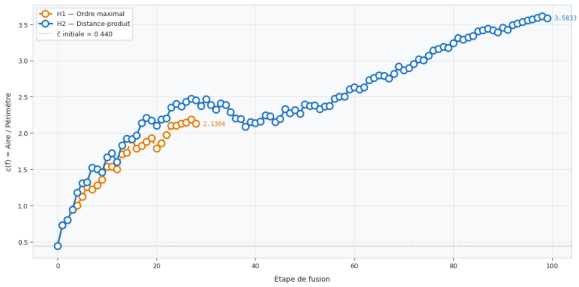


FIGURE 9 – Evolution of compactness with beam search, 100 iterations, $k = 100$ branches

Increasing the number of allowed branches (by a factor of 10, from $k = 10$ to $k = 100$) made it possible to generate only one additional complete solution, at the cost of five additional computation steps. This marginal gain demonstrates that, in the long run, the use of beam search for H_1 is counter-productive, given the prohibitive increase in execution time.

Intermediate conclusion. Consequently, for the remainder of our work, we will focus exclusively on the heuristic H_2 as a comparative baseline, since it offers the best compromise between algorithmic efficiency and quality of results. H_3 will be introduced as an empirical benchmark in the next section.

6 Comparative analysis and introduction of the empirical benchmark (H_3)

6.1 Limits of the initial comparative approach

One of the major limitations of our analysis lies in the fact that, until now, the performance of the heuristics has only been evaluated in a relative manner: H_1 against H_2. We therefore have no measure of their effectiveness with respect to a fixed and immutable standard.

Since the objective is to obtain the most compact configuration possible, the introduction of a benchmark is necessary. This makes it possible to quantify the real performance of our algorithms by evaluating the gap between the generated solutions and a theoretical optimum of compactness.

6.2 Definition of Heuristic 3 (H_3)

In anticipation of the study of larger instances, we introduce a third heuristic (H_3). It combines the rigor of the brute-force method with the efficiency of a greedy algorithm, with the aim of converging toward a final solution close to the optimum.

Definition 6.1 (Heuristic H_3)**.** *H_3 maintains a list of geometric shapes sorted in decreasing order of compactness. At each step:*

1. *The two most compact blocks f_1 and f_2 are selected.*
2. *All admissible fusion configurations between f_1 and f_2 are evaluated (Cartesian product of all tip–hollow pairs, over the 4 rotations).*
3. *The configuration maximizing the compactness index $c(f_1 \cup f_2)$ is retained.*

Unlike H_2, which filters candidates according to structural criteria (I, I'), H_3 performs no prior filtering: it exhaustively evaluates all fusion options between the two selected blocks. It is this local exhaustive character that gives it its precision, at the cost of increased complexity.

6.3 Study of time performance: H_2 vs H_3

Before analyzing the quality of the solutions, it is crucial to observe the computational cost of this new approach.

Size (n)	Time H_2 (ms)	Time H_3 (ms)	Factor H_3/H_2
2	0.71	1.30	1.83
4	28.95	538.16	18.6
8	1,039.75	3,131.56	3.01
12	3,910.37	10,473.36	2.68
16	7,472.80	19,854.86	2.66
20	13,121.15	30,460.23	2.32
24	19,233.91	44,315.72	2.30

TABLE 2 – Comparison of the computation times of H_2 and H_3 (same experimental conditions as Table 1).

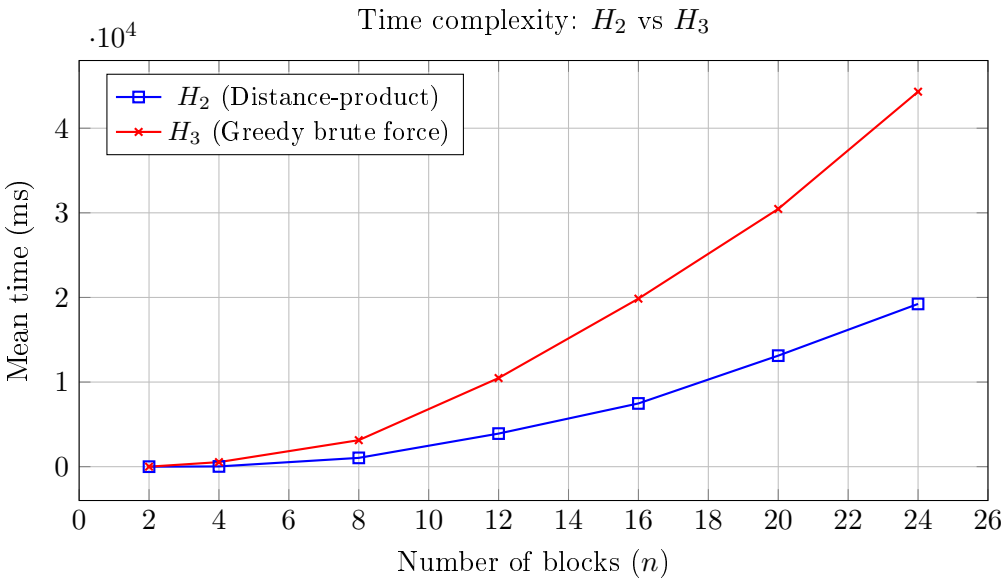


FIGURE 10 – Comparison of the computation times of H_2 and H_3 (data from Table 2). H_3 is about two to three times slower than H_2 for large instances, because of the exhaustive evaluation of the (tip, hollow) pairs.

6.4 Validation of H_3 against the theoretical optimum

6.4.1 Evaluation on small instances (exhaustive search)

In order to validate the relevance of H_3, we examine whether, for different instances of the problem, H_3 effectively converges toward the optimal-compactness solution (denoted here Optimum, obtained by exhaustive DFS).

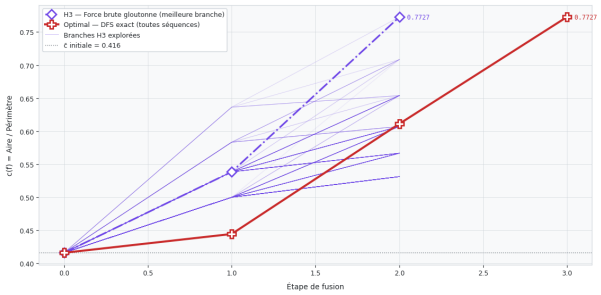


FIGURE 11 – Evolution of compactness for $n = 3$ blocks

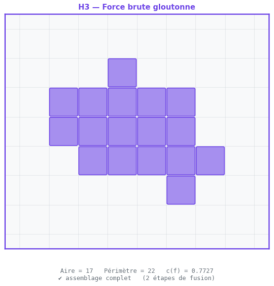


FIGURE 12 – H_3 shape, $n = 3$

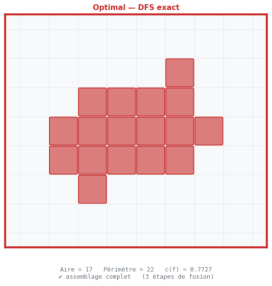


FIGURE 13 – Optimal shape, $n = 3$

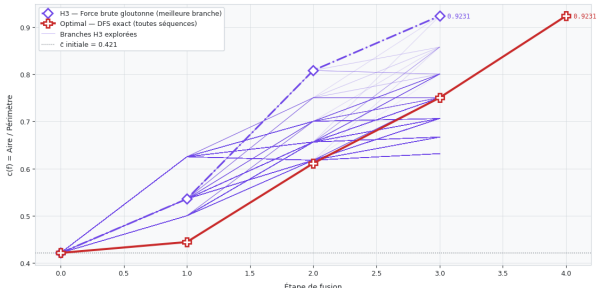


FIGURE 14 – Evolution of compactness for $n = 4$ blocks

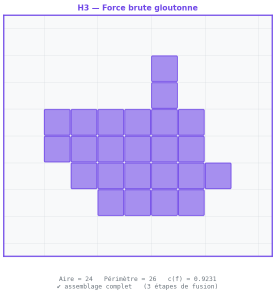


FIGURE 15 – H_3 shape, $n = 4$

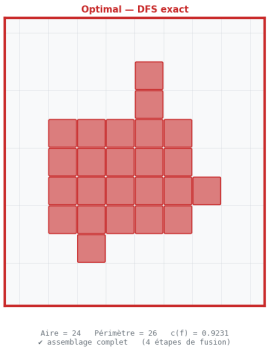


FIGURE 16 – Optimal shape, $n = 4$

From these first two tests, carried out on the smallest instance sizes, we observe that H_3 produces a complete solution with a compactness identical to that obtained by the exhaustive search (exact DFS). This validates H_3 as an **approximation of the optimum** for small instances.

6.4.2 Structural analysis and limits of the single metric

However, relying solely on compactness to evaluate the quality of solutions appears insufficient. Indeed, a visual analysis of the results highlights a structural difference between the solutions obtained, despite identical compactness values.

For an instance of size $n = 3$, a single block differs between the optimal solution and the one produced by H_3. For $n = 4$, this difference concerns two blocks. These gaps underline the need to introduce a second metric capable of capturing the morphological subtleties that compactness alone does not reveal. This motivation directly justifies the introduction of the resemblance metric in Section 7.

Remark 6.1. It should be noted that these results establish H_3 as an *approximation of the optimum* for small instances, and not as an absolute reference in the formal sense. For larger instances ($n \geq 5$), the convergence of H_3 toward the optimum is no longer guaranteed: it constitutes a practical upper bound rather than a theoretical bound.

6.5 Modeling the combinatorial complexity

H_3 constitutes a good approximation for small sizes. For large instances, exhaustive search becomes inapplicable. The total number of configurations evaluated by an exact algorithm can be modeled by:

$$\text{Total} = n! \times (p \cdot c \cdot k)^n$$

where:

— $n!$ is the number of permutations of the blocks (possible fusion orders),

— $p \cdot c$ is the number of possible tip–hollow pairs between two blocks,

— k is the number of possible relative placements per pair.

This super-exponential growth ($O(n! \cdot k^n)$) confirms that any exact approach is impracticable from $n \geq 10$, fully justifying the recourse to heuristics.

6.6 Conclusion on the H_3 approach and morphological results

In conclusion, although H_3 is more costly than H_2 (cf. Table 2 and Figure 10), it serves as a robust benchmark for moderate-size instances. For large instances, H_2 offers a better compromise.

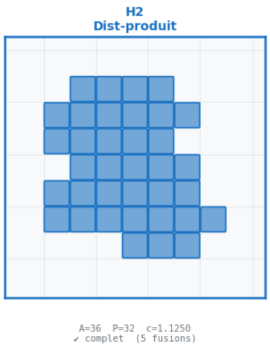


FIGURE 17 – H_2, $n =$ 6 blocks

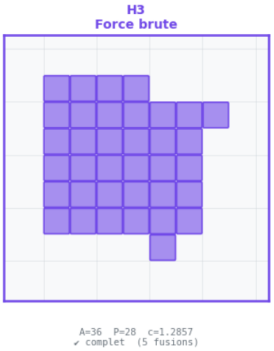


FIGURE 18 – H_3, $n =$ 6 blocks

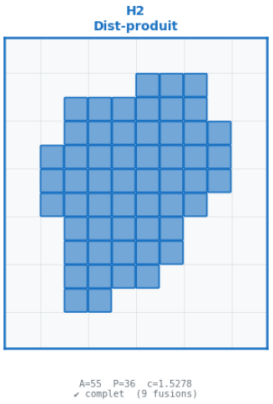


FIGURE 19 – H_2, $n =$ 10 blocks

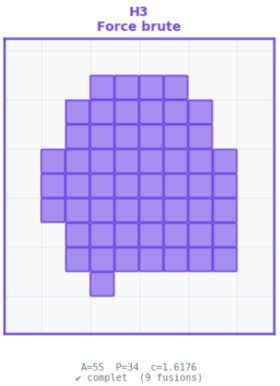


FIGURE 20 – H_3, $n =$ 10 blocks

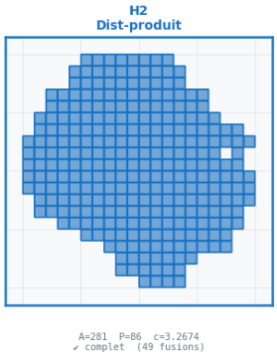


FIGURE 21 – H_2, $n =$ 50 blocks

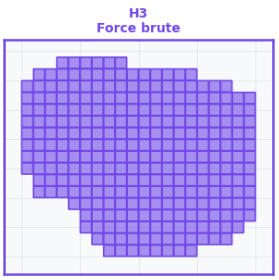


FIGURE 22 – H_3, $n =$ 50 blocks

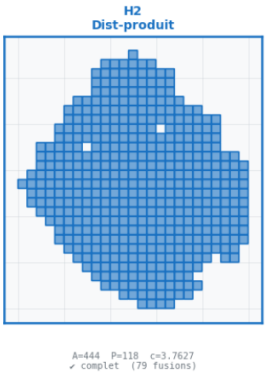


FIGURE 23 – H_2, $n =$ 80 blocks

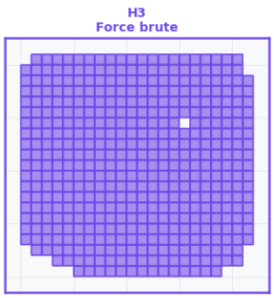


FIGURE 24 – H_3, $n =$ 80 blocks

The observation of the results shows that, as the number of blocks to merge increases, the morphology resulting from H_2 adopts a distinctly more "organic" appearance than that produced by H_3. Indeed, the structures generated by H_3 tend to reproduce conventional geometric shapes, almost exclusively square or rectangular.

This observation is consistent with the theoretical results (Corollary 2.1): the square is the optimal shape among rectangles in a discrete space because, for an equivalent area, it minimizes the exposed perimeter. H_3, by systematically seeking the fusion maximizing compactness, is naturally attracted toward these optimal shapes.

7 Structural characterization: Introduction of a complementary metric

7.1 Limits of compactness and need for a new indicator

As pointed out in the previous section (see Section 6.4.2), compactness, although essential, proves insufficient to differentiate two geometric shapes with identical scalar properties but distinct topologies. Two assemblies can indeed have strictly equal compactness while diverging radically in their spatial organization — this is precisely the case observed between H_3 and the optimum on the instances $n = 3$ and $n = 4$.

To overcome this limitation, we introduce a **resemblance metric** capable of quantifying the morphological divergence between two solutions in a normalized way.

7.2 Definition of the simple resemblance metric

Definition 7.1 (Simple resemblance metric)**.** *Let A and B be two blocks (after optimal alignment by rotation and translation minimizing the differences). Let N_{diff} be the cardinality of the symmetric difference $A \triangle B = (A \setminus B) \cup (B \setminus A)$, and $A_{tot} = |A \cup B|$ the area of their union. The simple resemblance metric is:*

$$R = \frac{N_{diff}}{A_{tot}} = \frac{|A \triangle B|}{|A \cup B|}$$

Remark 7.1. *$R \in [0, 1]$ is a normalized dimensionless measure:*

- *$R = 0$ means that the two shapes are identical (after alignment).*
- *$R = 1$ means that they have no cell in common.*

This normalization by A_{tot} rather than by $|A|$ or $|B|$ alone is important: it guarantees a symmetric measure that is not biased by the relative size of the shapes.

7.3 Spatial alignment methodology

The comparison of two blocks A and B (notably to compute the gap to the optimum H_3) requires an optimal alignment. Since blocks are invariant under rotation and translation, we use a symmetric-difference minimization protocol:

- **Normalization:** Blocks are brought back to their local origin ($\min_x = 0$, $\min_y = 0$).
- **Rotational exploration:** The four discrete rotation variants of B are generated (0°, 90°, 180°, 270°).
- **Optimization:** For each rotation, the symmetric difference $A \triangle B$ is computed.
- **Selection criterion:** The retained configuration is the one that minimizes the cardinality $|A \triangle B|$.

This method ensures that the resemblance metric R reflects the intrinsic structural divergence and not a mere spatial offset.

7.4 Evolution of the metric: Introduction of the eccentricity degree

The initial principle of the resemblance metric rests on a uniformity hypothesis: each divergent primary unit is assigned an identical weight, regardless of its position relative to the reference structure. However, this approach does not make it possible to distinguish a divergent unit adjacent to the contour of the optimal shape from an isolated or geometrically distant unit.

7.4.1 Concept of spatial weighting

From a morphological point of view, a shape whose excess units are adjacent to the walls of the optimal shape is considered structurally closer to the optimum than a structure exhibiting a large spatial gap between its mass and its extremities. To capture this nuance, we introduce an **eccentricity degree** w_i for each divergent cell i.

Definition 7.2 (Eccentricity degree)**.** *For a divergent cell i (belonging to $A \triangle B$), its eccentricity degree w_i is defined as its Manhattan distance to the perimeter of the reference shape A:*

$$w_i = d_{\text{Manhattan}}(i, \partial A) + 1$$

where ∂A denotes the set of perimeter cells of A, and the term $+1$ guarantees $w_i \geq 1$ for every divergent cell (even an adjacent one).

The farther a divergent cell is from the perimeter of the reference shape, the higher its weight w_i, and the greater its impact on the degradation of the metric. This mechanism severely penalizes shapes that "overflow" far from the optimum rather than those that differ only at the surface.

7.4.2 Formalization of the weighted metric

Definition 7.3 (Weighted resemblance metric)**.** *The weighted resemblance metric is defined by:*

$$R_w = \frac{\sum_{i=1}^{N_{diff}} w_i}{A_{tot}}$$

where N_{diff}, w_i and A_{tot} are defined as previously.

Remark 7.2. *Since $w_i \geq 1$ for all i, we always have $R_w \geq R$ (the weighted metric cannot be lower than the simple metric). The gap $R_w - R$ measures the "structural depth" of the divergences: a small gap indicates that the divergent cells are close to the reference surface, whereas a large gap reveals significant topological deformations.*

Remark 7.3. *It should be specified that this metric is methodologically relevant only in a comparative framework. It is designed as a discrimination tool making it possible to separate several approaches when they produce assemblies on identical instances. It does not provide an absolute measure of performance, but an indicator of relative convergence toward a reference standard (here, H_3).*

7.5 Comparative analysis by resemblance metric

7.5.1 Need for a comparative basis

Although the heuristic H_1 was set aside for the advanced phases because of its limitations (see Section 5), it retains here a usefulness as a morphological comparison point with respect to H_2. The figures below illustrate the morphological confrontation between H_1, H_2 and the reference H_3.

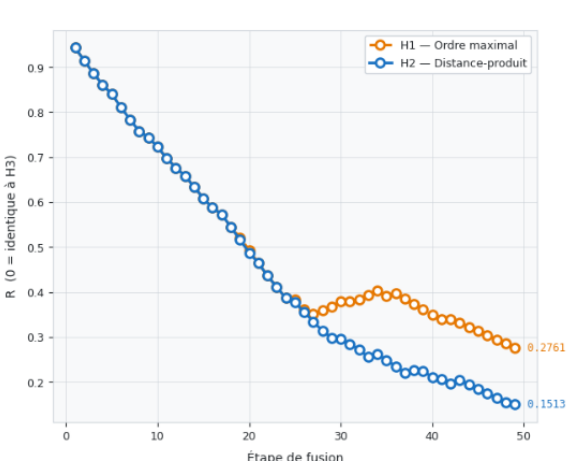


FIGURE 25 – Evolution of the simple resemblance metric R

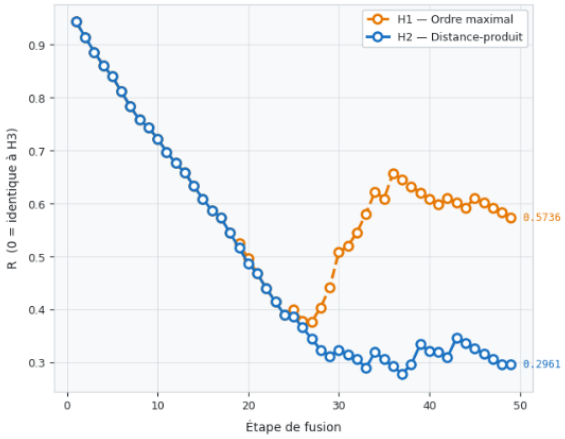


FIGURE 26 – Evolution of the weighted resemblance metric R_w

7.5.2 Interpretation of the experimental results

Phase 1 — Common convergence (steps 0 to 20). Over this portion, H_1 and H_2 follow *identical* trajectories. The low initial complexity of the shapes reduces the space of admissible candidates, forcing both heuristics to make identical fusion decisions. This uniform phase validates the consistency of the two approaches on simple configurations.

Phase 2 — Structural divergence (steps 20 to end). As the complexity of the shapes grows, the two heuristics diverge:

— H_2 maintains a regular convergence toward H_3 (decreasing values of R and R_w), which confirms that its fusion decisions are structurally closer to those of H_3.

— H_1 undergoes a sudden rise at steps 28–35, then stabilizes at a high level. This reveals that H_1 performs, at this stage, fusions that are locally compact but *structurally divergent* with respect to H_3.

7.5.3 Analysis of the impact of weighting

Heuristic	**Simple metric R**	**Weighted metric R_w**
H_1 (end of assembly)	0.5736	0.2761
H_2 (end of assembly)	0.2961	0.1513

TABLE 3 – Morphological divergence scores at the end of assembly, with respect to H_3.

The systematic drop in scores with the weighted metric (factor ≈ 2 in both cases) confirms that the divergent cells remain *close to the perimeter* of H_3. In other words:

— The errors of H_1 are **surface deformations** rather than major spatial aberrations — even if its overall shape remains stuck far from the optimum.

— H_2 converges structurally toward H_3, with a final divergence almost two times smaller than H_1 according to both metrics.

7.6 Summary of the morphological analysis

The introduction of the resemblance metrics, and more particularly of the weighted metric R_w, marks a paradigm shift: it is no longer only a matter of evaluating the scalar efficiency of a shape, but of quantifying its morphological fidelity with respect to a reference standard. The experiment brought two fundamental points to light:

— **The structural superiority of H_2:** Unlike H_1, the heuristic H_2 exhibits a regular convergence and a final proximity to H_3 that is distinctly more pronounced. The results of Table 3 confirm that H_2 halves the gap to the optimum compared with H_1.

— **The nature of the divergences:** The study of the eccentricity degree reveals that the observed gaps remain essentially "surface deformations". The fact that the weighted scores remain proportional to the simple scores suggests that the divergent cells gravitate around the perimeter of H_3 without creating major spatial aberrations.

In conclusion, the metric R_w asserts itself as a robust discrimination tool to validate the algorithmic choices of H_2 regarding the preservation of the overall integrity of the shape. This analysis must however be conducted jointly with the simple metric R: whereas the latter quantifies the raw volume of divergent primary units, R_w makes it possible to evaluate their "structural depth", thus offering a finer reading of the spatial organization of the block.

8 Discussion and conclusion

8.1 General conclusion

This study made it possible to formalize and compare three heuristic approaches to the problem of the automatic assembly of discrete geometric shapes.

The heuristic H_1, based on the alignment of maximal-order vertices, demonstrated remarkable speed ($O(V + N)$), making it suitable for applications requiring real-time processing. However, its sensitivity to morphological complexity limits its use when structural precision is required. In contrast, the heuristic H_2, through its exhaustive analysis of dimensional compatibilities and its multi-criteria selection (I, I'), established itself as a robust solution, capable of converging significantly toward the optimum represented by H_3.

One of the major contributions of this work lies in the introduction of a **resemblance metric weighted by eccentricity**. This tool proved its ability to finely quantify the quality of a fusion beyond the simple area, by penalizing structural aberrations according to their distance from the optimal shape. This metric, whose construction is inspired by the Jaccard distance but enriched with an original spatial weighting, constitutes a contribution that is reusable independently of the assembly framework presented here.

In conclusion, the duality between compactness and morphological resemblance offers a complete analysis framework for discrete optimization. Research perspectives are now oriented toward the parallelization of collision tests and the application of this model to generative design and 3D packing problems, where the management of topological complexity remains a major challenge.

8.2 Study limitations and research perspectives

Although the results obtained validate the relevance of the morphological approach, several limitations intrinsic to the current model must be highlighted.

8.2.1 Combinatorial explosion and scalability

The time analysis highlighted the quadratic nature of the heuristic H_2. While the latter performs well for moderate-size instances ($n \leq 24$), moving to complex systems involving hundreds of unit blocks poses a scalability challenge. The growth in the number of vertices V induces an exponential increase in the candidate pairs to evaluate in the set I. An optimization of the data structures, notably through the use of spatial partitions or dynamic bounding boxes, would be a necessary improvement path to maintain acceptable computation times.

8.2.2 Statistical robustness of the experimental measurements

The performance measurements presented rely on 5 repetitions per configuration. This sample size is sufficient to illustrate asymptotic trends, but insufficient for formal statistical conclusions. A later study will have to increase this number to a minimum of 30 repetitions, accompanied by Wilcoxon tests or paired t-tests to validate the comparisons between heuristics.

8.2.3 Dependence on spatial discretization

The study relies exclusively on a discrete space (regular grid). Although this framework is suited to many industrial applications, it imposes constraints on rotation (90° increments) and positioning that simplify the topology of the vertices. Extending the model to a continuous environment or to unstructured meshes would require a complete redefinition of the notion of vertex order and of the associated compactness metric.

8.2.4 Locality of heuristic decisions

Finally, H_1 and H_2 remain greedy-type strategies. They operate through locally optimal choices at each fusion step. Yet, as the divergence of H_1 in the advanced phase has shown, a locally compact choice can lead to a global morphological dead end. The integration of backtracking mechanisms or the use of metaheuristic algorithms (tabu search, simulated annealing [2]) would make it possible to explore solutions that the irreversible nature of the current fusion hides.

Annexes

A.1 Proof of Proposition 2.1

Proof by successive area removal. Consider a full rectangle E of dimensions (m, n). Its compactness is:

$$C_E = \frac{m \cdot n}{2(m+n)}$$

Let b be a connected block obtained by removing one unit of area from E ($\text{Area}_b = m \cdot n - 1$). We distinguish three cases for the change of the perimeter P_b relative to P_E:

1. **Removal at a corner:** The perimeter is preserved, $P_b = P_E$.
2. **Removal on a lateral segment:** The perimeter increases, $P_b = P_E + 2$.
3. **Removal inside the block:** The perimeter increases, $P_b = P_E + 4$.

In all configurations, $P_b \geq P_E$. Since $\text{Area}_b < \text{Area}_E$, we have:

$$C_b = \frac{m \cdot n - 1}{P_b} \leq \frac{m \cdot n - 1}{P_E} < \frac{m \cdot n}{P_E} = C_E$$

By induction on the number of removed units, this result extends to any block of area $m \cdot n - k$ with $k \in \mathbb{N}^*$. By transitivity, the full rectangle E maximizes compactness among all connected blocks whose bounding box is exactly (m, n). □

A.2 Proof of Proposition 2.2

Proof. We show that among all full rectangles of area n^2, the square of side n maximizes compactness.

Lemma 8.1. *Every rectangle of area n^2 can be written in the form $(a, b) = \left(\frac{n^2}{p}, p\right)$ where p is a positive divisor of n^2.*

Proof of the lemma. Let (a, b) be a rectangle of area n^2 with $a, b \in \mathbb{N}^*$. Then $a \cdot b = n^2$, which implies that b divides n^2. Setting $p = b$, we obtain $a = \frac{n^2}{p}$, hence the announced form.

Proof of the proposition:

The compactness of the square is $c_{opt} = \frac{n^2}{4n} = \frac{n}{4}$.

The compactness of an arbitrary rectangle (a, b) of area n^2 is:

$$c = \frac{n^2}{2(a+b)}$$

We want to show that $c_{opt} \geq c$, i.e., $\frac{n}{4} \geq \frac{n^2}{2(a+b)}$, which is equivalent to $a + b \geq 2n$.

Now, by the arithmetic–geometric mean inequality (AM-GM):

$$\frac{a+b}{2} \geq \sqrt{ab} = \sqrt{n^2} = n \implies a + b \geq 2n$$

with equality if and only if $a = b = n$, i.e., the square. Hence the square maximizes compactness among all rectangles of area n^2. □

A.3 Proof of Proposition 2.3

Proof. Let b be a block of area A_b and perimeter P. By definition, its compactness is $c(b) = A_b/P$.

1. For any polyomino of perimeter P and area A_b, we have the discrete isoperimetric inequality $P \geq 4\lceil\sqrt{A_b}\rceil$, and in particular $P \geq 4\sqrt{A_b}$.
2. We deduce: $c(b) = \dfrac{A_b}{P} \leq \dfrac{A_b}{4\sqrt{A_b}} = \dfrac{\sqrt{A_b}}{4}$.
3. By hypothesis, $A_b \leq n^2$, hence $\sqrt{A_b} \leq n$.
4. Substituting, we obtain $c(b) \leq \dfrac{n}{4}$.
5. Now, for the square S_n of side n, the area is n^2 and the perimeter is $4n$, hence $c(S_n) = \dfrac{n^2}{4n} = \dfrac{n}{4}$.

Thus, $c(b) \leq c(S_n)$ for every block of area less than or equal to n^2. □

A.4 Proof of Proposition 3.1

Proof. The admissibility of an interlocking rests on the coincidence of the boundaries of the segments adjacent to the vertices p and q. For each case of Proposition 3.1, the geometric condition is that the translation δ applied to f_2 to align q with p does not create an intersection between the interiors of f_1 and f_2.

— **Case 1:** If all the neighbors of p and q are respectively tips and hollows, the contours fit together perfectly. No additional constraint is necessary.

— **Case 2:** If a neighbor q^+ of q is a tip, it indicates a "protuberance" of f_2 toward which f_1 presents a protrusion. The condition $\min(d(p^-, p), d(p, p^+)) \leq d(q, q^+)$ ensures that the protrusion of f_1 is narrow enough not to exceed the available void on the side of q^+. If this condition is violated, the segment of f_1 overflows into the region occupied by f_2, creating an overlap.

— **Case 3:** Symmetric to Case 2, with the roles of f_1 and f_2 exchanged. The condition $d(p, p^+) \leq \max(d(q^-, q), d(q, q^+))$ guarantees that the protrusion of f_2 fits within the internal space of f_1.

— **Case 4:** The conjunction of the conditions of Cases 2 and 3 is necessary and sufficient: each of the two protrusions must independently satisfy its non-overlap criterion, and both conditions must hold simultaneously because the constraints on each side are independent.

□

A.5 Proof of Proposition 3.2

Proof. Let $\Pi = d_1 \cdot d_2$ be the product of the lengths of the facets adjacent to the tip p, representing the local area of influence of the vertex. Suppose that the hollow q exhibits an axial symmetry along one of the orthogonal axes of the discrete plane (explicit hypothesis of the proposition).

Under this hypothesis, the lengths of the segments of the hollow q in the two orthogonal directions are interchangeable without modifying the zone of influence. If $d_1 \cdot d_2 = d'_1 \cdot d'_2$, then the redistribution $(d_1, d_2) \to (d'_1, d'_2)$ preserves the area of influence Π and, by the symmetry of q, does not modify the non-overlap conditions established by Proposition 3.1.

In the discrete space, the validity of the interlocking is governed by a system of translations $x + \delta_x = x_q$ and $y + \delta_y = y_q$. The permutation of the factors $d_1 \leftrightarrow d'_1$ corresponds to a

horizontal/vertical redistribution of the lengths which, under the hollow-symmetry constraint, produces an admissible translation vector (δ'_x, δ'_y). The admissibility of p' with respect to q is therefore guaranteed, provided that the decomposition (d'_1, d'_2) respects the distance constraints of Cases 2–4 of Proposition 3.1.

Note. In the absence of symmetry of the hollow q, the permutation of the factors may produce an overlap. Verification in the general case requires an explicit non-overlap test. □

Bibliography

References

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[2] Kirkpatrick, S., Gelatt, C. D., & Vecchi, M. P. (1983). Optimization by simulated annealing. *Science*, 220(4598), 671–680.